Report

Indigenous and local knowledge dialogue workshop

on the

IPBES assessment

of

spatial planning and ecological connectivity

18 – 22 August 2025, Bardiya, Nepal

Suggested citation

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1 Introduction

This report summarizes the proceedings from the first Indigenous and local knowledge (ILK) dialogue workshop for the IPBES assessment of spatial planning and ecological connectivity that is being developed by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystems Services (IPBES).

The dialogue workshop was held in Bardiya, Nepal from 18 to 22 August 2025. It aimed to provide a platform for discussion between Indigenous Peoples and local communities and assessment authors, with a focus on key ILK concepts, themes, questions, challenges, opportunities, resources and other issues relating to the assessment.

The workshop also discussed the IPBES assessment on monitoring biodiversity and nature's contributions to people (the "monitoring assessment"), and constituted the second workshop for that assessment. That part of the workshop is the subject of a separate report.

This report aims to provide a written record of the dialogue workshop as it related to the assessment of spatial planning and ecological connectivity, so that it can be used by assessment authors to inform their work on the assessment, and by all dialogue participants who may wish to review and contribute to the work of the assessment moving forward, as well as others who may be interested in subjects relating to spatial planning and ecological connectivity and ILK.

The report is not intended to be comprehensive or to provide definitive resolution to the many engaging discussions that emerged during the workshop. Rather, it serves as a written record of those discussions, which will continue to develop and evolve in the months and years ahead. For this reason, clear points of agreement are discussed, but also, if there were diverging views among participants, these are also presented for further attention and discussion.

The text in section 3 represents an attempt to reflect solely the views and contributions of the participants in the dialogue. As such, it does not represent the views of IPBES or UNESCO or reflect upon their official positions.

The agenda and participants' list for the dialogue workshop are provided in annexes 1 and 3.

2 Background

2.1 The IPBES assessment of spatial planning and ecological connectivity

2.1.1 Structure

The assessment of spatial planning and ecological connectivity (full title: "the methodological assessment of integrated biodiversity-inclusive spatial planning and ecological connectivity") commenced in 2025 and will be considered by the Plenary[1] at its thirteenth session, which is expected to be held in 2026.

It is a "fast-track" assessment, which means that it has only one external review period and is conducted over a shorter period of time (as opposed to the full IPBES assessments which generally take three to four years and have two external review periods).

The assessment's scoping report is available here.

The assessment report will consist of a summary for policymakers and six chapters. The chapters are as follows:

1. Setting the scene
2. Implementing target 1 of the Kunming-Montreal Global Biodiversity Framework (KMGBF)[2] on biodiversity-inclusive spatial planning
3. Implementing targets 2 and 3 of the KMGBF, on restoration and protected areas and other area-based conservation measures
4. Maintaining, restoring and enhancing ecological connectivity
5. Spatial planning for the future

[1] The Plenary is the body through which States that are members of IPBES take decisions, which usually meets around once a year.

[2] The Kunming-Montreal Global Biodiversity Framework was adopted during the fifteenth meeting of the Conference of the Parties (COP 15) of the CBD. This framework sets out an ambitious pathway to reach the global vision of a world living in harmony with nature by 2050. Among the Framework's key elements are 4 goals for 2050 and 23 targets for 2030.

6. Creating an enabling environment for integrated biodiversity-inclusive spatial planning and ecological connectivity

2.1.2 Scope and rationale

The assessment is a *methodological assessment*, which means that it assesses available methodologies for addressing a specific theme.

The assessment will:

- Provide options for avoiding land and sea use change that negatively affects biodiversity;
- Provide options for improving planning for effective conservation, restoration and sustainable use of nature and its contributions to people, including through protected areas and other effective area-based conservation measures;
- Cover methods and tools for integrating biodiversity and promoting connectivity in spatial planning;
- Cover lessons learned and best practices on how ecological connectivity contributes to biodiversity conservation, restoration, sustainable use and management, as in the case of migratory species, for example;
- Look at participatory approaches to spatial planning, including those involving Indigenous Peoples and local communities; and
- Illustrate the potential of spatial planning to simultaneously achieve a range of global goals, particularly those related to biodiversity, food, poverty, water, health and climate change.

2.2 Context for the dialogue workshop

2.2.1 IPBES and ILK

IPBES is an independent intergovernmental body established to strengthen the science-policy interface for biodiversity and ecosystem services for the conservation and sustainable use of biodiversity, long-term human well-being and sustainable development.

Since its inception in 2012, IPBES has recognized that Indigenous Peoples and local communities possess detailed knowledge of biodiversity and ecosystem trends. In its first work programme (2014-2018), IPBES built on this recognition through deliverable 1 (c), *Procedures, approaches, and participatory processes for working with Indigenous and local knowledge systems.* The IPBES rolling work programme up to 2030 includes objective 3 (b), *Enhanced recognition of and work* *with Indigenous and local knowledge systems*, which aims to further this work. The IPBES conceptual framework also contains explicit recognition of diverse knowledge and value systems.

Recognizing the importance of ILK to the conservation and sustainable use of ecosystems as a cross-cutting issue relevant to all of its activities, and noting also that approaches and methods for working with ILK and Indigenous Peoples and local communities in global- and regional-scale assessments would need to be developed, the IPBES Plenary established a task force on ILK systems and agreed on terms of reference guiding its operations towards implementing this deliverable. IPBES' work with Indigenous Peoples and local communities and on ILK is supported by a technical support unit for ILK, hosted by UNESCO.

Key activities and deliverables of the task force and technical support unit on ILK so far include:

- Progress in the development of approaches and methodologies for working with ILK was made during previous IPBES assessments (Pollination, Pollinators and Food Production, Land Degradation and Restoration, four Regional Assessments and a Global Assessment of Biodiversity and Ecosystem Services, Sustainable Use of Wild Species, Diverse Values and Valuation of Nature, Invasive Alien Species, the Interlinkages among Biodiversity, Water, Food and Health, and Transformative Change);
- The development and implementation of the "approach to recognizing and working with ILK in IPBES", which was formally approved by the Plenary at its fifth session in 2017 in decision IPBES-5/1, which sets out principles and approaches for IPBES's work with ILK;
- Development and implementation of methodological guidance for recognizing and working with ILK in IPBES, which aims to provide further detail and guidelines on how to work with ILK within the IPBES context; and
- Development and implementation of a "participatory mechanism", a series of activities and pathways to facilitate the participation of Indigenous Peoples and local communities in IPBES assessments and other activities, which includes organizing ILK dialogue workshops for the IPBES assessments.

2.2.2 Working with ILK in the assessment process

Following the IPBES approach to ILK and as part of the participatory mechanism, two ILK dialogue workshops are held during the cycle of a "fast-track" IPBES assessment, as follows:

- Discussing key ILK themes and framing of the assessment (in the first months after an assessment commences); and
- Reviewing the first draft of the chapters and summary for policymakers (during the assessment's external review period).

These workshops bring together members of Indigenous Peoples and local communities, Indigenous and non-indigenous scientists, and authors of the assessment to discuss key themes relating to the assessment. They are part of a series of complementary activities for working with ILK and enhancing participation by Indigenous Peoples and local communities throughout the assessment process.

Other activities during an assessment include an online call for contributions, invitations to contributing authors and review of diverse literature and materials.

2.3 Objectives of the ILK dialogue workshop

The first ILK dialogue aims to provide a platform for discussion between participants of the workshop about potential key ILK concepts, themes, questions, challenges, opportunities, resources and other issues relating to the IPBES assessment of spatial planning and ecological connectivity. Specific aims of the dialogue include:

- Developing recommendations from Indigenous Peoples and local communities for key themes or questions of the assessment, which will help to shape an ILK narrative for the assessment and direct the collection, synthesis and analysis of information;
- Discussing how Indigenous Peoples and local communities conceptualize, use, manage and govern landscapes and seascapes;
- Discussing challenges, risks and opportunities related to the assessment from the perspectives of Indigenous Peoples and local communities;
- Beginning to develop case studies of relevance to the assessment;
- Identifying experts who can contribute to the assessment as contributing authors or participants in future dialogue workshops and review processes; and
- Identifying resources and sources of information that could be included in the assessment.

2.4 Methods for the dialogue workshop

The workshop was held in-person over five days, and it first addressed the assessment of spatial planning and ecological connectivity before turning to the assessment of monitoring biodiversity, although links and connections between the two assessments were drawn and emphasized throughout. Time was also set aside at the beginning of the workshop to allow participants to discuss the issues they wished to address at the workshop, and how these issues should be approached. The agenda is presented in annex 1 of this report.

The process for the dialogue workshop included:

- Initial presentations and discussions on:
    - IPBES and its goals and methods;
    - Workshop aims, methods, and free, prior and informed consent (FPIC); and
    - The assessment, its goals and proposed methods;
- Discussions around the key ILK questions and themes for the assessment as a whole and for each chapter of the assessment. Participants were asked to reflect on key questions and themes and provide examples and case studies that could support work in the chapters;
- A day of community visits to the Tharu community in Dalla, Bardiya, including a community meeting and visits and discussions around the community; and
- Two sessions of the Indigenous and local community caucus, an informal group composed of Indigenous and local community representatives, with ad hoc rules of engagement, to give participants a space to discuss any questions, concerns or issues in relation to IPBES, the assessment and the workshop.

2.5 Free, prior and informed consent

Free, prior and informed consent (FPIC) principles are central to IPBES work with Indigenous Peoples and local communities. A series of ethical principles have been developed to ensure that FPIC is followed in IPBES activities. These principles were agreed upon by the participants from Indigenous Peoples and local communities and IPBES authors in the dialogue, recognizing that participants, authors and the IPBES technical support units have different responsibilities within the process. The principles will be followed by participants from Indigenous Peoples and local communities, the assessment authors group and the IPBES technical support units. The full agreed-upon text and the names of those agreeing to these principles are provided in annexes 2 and 3 to this report.

2.6 Benefits to Indigenous Peoples and local communities of participating in the assessment and other activities

During previous ILK workshops, participants emphasized the need for clear benefits to Indigenous Peoples and local communities from their involvement and participation in an assessment process. It was noted that IPBES does not benefit financially from its processes or products, and that the main products of IPBES are publicly available materials, including assessment reports, their summaries for policymakers, webinars and other resources, to provide free and reliable information for policymakers and decision-makers and actors at all levels, including Indigenous

Peoples and local communities. Key benefits discussed for Indigenous Peoples and local communities from participating in dialogue workshops and the overall assessment process include:

- The opportunity to enhance relationships, and share experiences, examples and cases among Indigenous Peoples and local communities around the world;
- The opportunity for Indigenous Peoples and local communities to engage in dialogue around experiences and knowledge with IPBES assessment authors;
- The opportunity to bring ILK and the concerns and visions of Indigenous Peoples and local communities to the attention of policymakers and decision-makers; and
- Use of the final assessments as tools to support Indigenous Peoples and local communities when engaging with policymakers, decision-makers and scientists.

Part of the planning for the final assessment includes the development of an accessible summary specifically for Indigenous Peoples and local communities, as well as webinars that present the results to Indigenous Peoples and local communities.

Figure 1: Participants during the field visit looking across to Bardiya National Park.

3 Spatial planning and ecological connectivity: Key recommendations and learning from the dialogue workshop[3]

Throughout the workshop, participants engaged in discussions about various aspects of the assessment of spatial planning and ecological connectivity. This section highlights the key messages, recommendations and examples shared by Indigenous Peoples and local communities during the workshop. To preserve authenticity, the content presented here closely reflects the participant´s statements and comments during the workshop, with only minor editorial adjustments for clarity.

3.1 Aspirations

Participants noted that the spatial planning assessment is potentially highly important for Indigenous Peoples and local communities, as it could provide an opportunity to rethink the environmental and socio-cultural impacts of traditional conservation approaches and how protected areas are established and managed, as these often impact Indigenous and human rights. The assessment could explore ways to prevent conflicts, promote the inclusion of Indigenous Peoples and local communities in spatial planning processes, and support "rights-based" approaches. It can also help to set a road map for the implementation and achievement of crucial targets of the KMGBF. These targets hold potential to enhance biodiversity conservation and respect for Indigenous rights and human wellbeing, but they also have the potential for negative impacts if implemented without the necessary consideration, understanding and processes.

[3] Disclaimer: The text in section 3 represents an attempt to reflect solely the views and contributions of the participants in the dialogue. As such, it does not represent the views of IPBES or UNESCO or reflect upon their official positions.

3.2 Conceptualizations

Participants agreed that it is first important to understand how Indigenous Peoples and local communities conceptualize and understand landscapes, seascapes and connectivity.

3.2.1 Conceptualizations of landscapes and seascapes

Many participants emphasized that Indigenous Peoples often see landscapes and seascapes as interwoven biocultural spaces, in which people are inseparable from nature. The concept of "territory" is also used to express the interwoven relationships between lands, waters, people, livelihoods, culture, identity, knowledge and myriad other connections.

Another participant noted that some Indigenous Peoples may not have a concept of "landscape", as such, as they do not separate the cultural and social elements from the lands on which they live. Understanding Indigenous concepts of "landscape" may therefore need a broader view encompassing elements and interconnections beyond what might be considered a landscape by others in society.

Participants from Russia and New Zealand also shared their conceptualization that landscapes are infused with spirituality, and spiritual relationships exist above all others, and that this creates obligations and responsibilities for how people act within a landscape and seascape. Spirituality was similarly highlighted as key by participants from Canada, Mexico and Peru. For example, in Mexico some large animals such as jaguars are protected because of their spiritual significance. Similarly, participants from Nepal also explained the importance of sacred landscapes, including the Himalayas.

A participant from Russia highlighted that for Indigenous Peoples, the land does not belong to people, but instead people belong to the land, and that this impacts how they understand management and planning within it.

Participants from Uganda and Mexico also noted that many Indigenous Peoples see themselves as holding the land in trust, both for ancestors and the unborn future generations. The landscape cannot only be defined by living generations. The need for intergenerational reciprocity means that people should protect their landscapes and seascapes. Many such concepts of reciprocity and responsibility between people and landscapes and seascapes and encoded in Indigenous and local languages, worldviews and philosophies.

3.2.2 Conceptualizations of connectivity

Participants highlighted that connectivity is key to many Indigenous and local worldviews and philosophies relating to landscapes and seascapes. In particular, they highlighted that within many of these conceptualizations, connectivity includes people, within biocultural systems that tie landscapes and seascapes to customary practices, management and governance, land tenure, food systems, health systems, livelihoods, trade and economy, spirituality, sacred sites and burial places.

A participant from Canada noted that Indigenous Peoples often see themselves as connected to all the elements in a landscape, including animals, plants, inanimate objects, spirits and past and future generations. A participant from Nepal also highlighted connections to the sky and planets as part of the community's vision of connectivity.

Participants from Canada, Mexico and Peru explained that peoples' identities are also often connected to the landscape or seascape. For example, in some parts of Canada, clan systems tie people to a specific animal or place. This helps to build connections between families and between people and the land, including protecting lands, waters and animals.

Participants from Nepal, Mexico and Peru also highlighted that connectivity between people and landscapes is often an ongoing practice among Indigenous Peoples, embedded in festivities, rituals and active guardianship of the environment (see examples from Nepal and Peru, below).

Similarly, participants from Nepal explained the importance of landscapes as interconnected with people by the ongoing tasks and activities that take place within them, for example *Khalihaan*, a private or community space where cultural ecological practices are carried out, including winnowing, threshing, funneling and storage of grains.

Participants also reflected on the ways that Indigenous Peoples and local communities understand the physical connections within landscapes, for example a participant from New Zealand shared the concept of *Ki uta ki tai – from the origins to the sea*, which explains the connectivity of the water system and ecosystems throughout a river system.

A participant from Canada also noted that concepts of connectivity are often embedded in Indigenous languages, and place names may include elements of connection, e.g., relating to bird migrations. Similarly, in East Asia, the words and concepts of *satoyama* or *satoumi* help to show the connections between paddy fields, farmlands and soils and the surrounding mountains, often through the river systems that also flow down to the ocean.

Participants also noted the need for conceptual shifts in broader society and science, to move away from compartmentalization and towards recognizing interrelationships and connections. "Connectivity" may therefore not only be conceived as connectivity between ecological elements, or it will continue to enhance processes of fragmentation and separation between people and between people and landscapes and seascapes. Participants also noted that connectivity must also include political dimensions – it is essential that different levels of government are connected across scales, and that these are also connected to local people and nature.

Examples

Local participants shared the ways that the Tharu people of Bardiya, **Nepal** are connected to their lands. The Tharus are mostly farmers and live close to the rivers, as they also rely on fishing, and fish are also important for festivals and ceremonies. They also use the forest, including for collecting mushrooms, ferns and herbs, and medicines that they can use to cure illnesses and snake bites. These medicinal plants also have spiritual elements that enhance their healing properties. Some of the Tharu people have animals as deities, such as elephants, snakes and tigers, which are worshipped throughout the year. At certain times of the year, the connections between people, animals, plants, water, land and spirits are celebrated through special rituals and offerings, including in the month of *Shrawan* (from 16 July to 16 August), where the connections between people, animals, plants and spirits are painted on the wall of a building (see figure 1 below). Moreover, sacred sites next to their fields are used to perform ceremonies related to the agricultural cycle, including ceremonies before rice is planted, during the growing season and once it is harvested. They also perform ceremonies to the spirits of animals, including wild boar, so that they will stay away from the fields and not damage the crops. The *Badhghar* customary governance of Tharu communities in Bardiya has been contributing to land and water management and sustainable use of natural resources since time immemorial. Roles and responsibilities in the community are decided during festivals by consensus, including who will maintain ceremonies and other spiritual activities.

Figure 2: Tharu ritual in which connections between people, nature and spirits are celebrated, in Bardiya, Nepal.

August in the highlands of **Peru** is a crucial time, as this is when Pachamama (Mother Earth) is most fertile, so the communities perform sunrise ceremonies and offer a *despacho* to Pachamama. A *despacho* is a ceremonial offering that includes symbolic items such as corn, dried fruit, flowers, coca leaves and sweets, along with other traditional foods, all bundled together as a gift. Each of these offerings is infused with heartfelt prayers and intentions, serving as an act of gratitude and reciprocity to Pachamama and the *Apus*, the mountain spirits. Through this practice, the Andean people of Peru honour the deep connection they share with nature and the spiritual realm.

In Thiruvananthapuram, Kerala and Kanyakumari district of Tamil Nadu in **India**, the Mukkuvar community in Thiruvananthapuram, Kerala, and Kanyakumari district of Tamil Nadu, use a navigational knowledge system known as *kanicam* to find locations on the ocean. This knowledge is based on in-depth knowledge of the ocean, and its interconnections with the land. As there are no GPS or similar devices, knowledge of mountains, rivers, land and the sky is needed to locate a place or an underwater ecosystem. For example, the mountains are reference points during the day, and knowledge-holders measure how tall they observe them to be from different points in the ocean, using a kind of triangulation. However, as most coastal Indigenous communities carry out livelihood practices at night, they also need an understanding of the stars and planets. The whole system is interconnected, for example rivers are connected to the sea, so the community does not conceptually separate them. If a river is polluted, the sea will also be polluted.

Figure 3: Sketch of how the Mukkavar people understand connections between land and sea (with thanks to Johnson Jament).

A participant from **Uganda** shared how, for pastoralists, landscapes are a web of grazing areas, wildlife areas, water points, homesteads, mobility routes and sacred spaces that people move through, following in depth-knowledge and connections to known places, and managed through customary governance processes (described below in section 3.3).

Figure 4: A sketch of a pastoralist landscape in Uganda (with thanks to Loupa Pius).

3.2.3 Language

Participants discussed the different ways that Indigenous and local languages can encode knowledge and concepts, particularly as they relate to landscapes, seascapes, planning and connectivity.

Participants highlighted that many place names and words are tied to landscapes and seascapes. Many places are named after their role in the ecosystem, and from that, people can know a specific place's role in the landscape. Another participant highlighted that it is not just a matter of individual words, but also the ways people talk and the metaphors they use to describe and explain their connections to landscapes and seascapes.

Example

In the Mukkuvar Indigenous language in **India**, there are over 1000 words related to the ocean, which is far more than in other languages spoken in the region. For example, there is recognition of two estuarine systems – *pozhi* (river never joins the sea) and *azhi* (river naturally joins the sea). This concept, which highlights the differences between these two environments, is not always well understood by science. Another term is *Kappal paru* – shipwrecked areas and other artificial reefs. *Kappal* means ship and *paru* means ecosystem, so the language explains that this is not just the wreck of a ship, it is an entire ecosystem.

A participant noted that for some Indigenous Peoples and local communities, their knowledge and connections to lands and waters may be more experiential and connected to living and doing, rather than connected to clearly defined concepts. Another participant noted that some communities communicate more through art or other modes of representing knowledge, and this should also be considered when engaging in efforts to understand community knowledge of landscapes and seascapes.

3.3 Land planning and decision-making by Indigenous Peoples and local communities

Participants highlighted that many Indigenous Peoples and local communities have long histories of successfully managing landscapes, seascapes and connectivity. Moreover, their more holistic conceptualizations of landscapes, seascapes and connectivity, described in section 3.2 above, often lead to broader benefits across biocultural systems, rather than focusing on narrow goals, for example, only focusing on biodiversity to the exclusion of human wellbeing. They also noted that it is important to keep in mind that some aspects of Indigenous and local ways of managing lands and waters, and their relationships with them, may not immediately look like spatial planning to people trained in academic disciplines. Nonetheless, they represent diverse ways of planning and managing lands and waters, and should be explored and acknowledged as such.

A participant from Mexico shared that values, spirituality and cosmologies are often foundations for making decisions around land use planning, and many communities organize their relationships with, and uses of, lands based on this. Many Indigenous Peoples and local communities also consider their ancestors, and the ways that they related to landscapes and seascapes, when making decisions about the use of their lands and waters, as these lessons can often be adapted to current situations and contexts. A participant from Canada also shared the principle of protecting and thinking of the next seven generations, and how this defines how people think about activities and stewardship of landscapes.

Participants also noted that community-imposed restrictions on use and access around spiritual sites are another method that Indigenous Peoples and local communities use to plan and manage their use and protection of lands and waters. Sometimes these restrictions are permanent, or they may be in place for certain periods of time. The continuation of spiritual practice is often essential for the continued conservation of these areas. Other systems of communal management include systems of farming on steep lands in the Andes using terraces.

Participants also noted that many Indigenous Peoples and local communities are also engaging in restoration of landscapes and seascapes, to combat the degradation that many are witnessing in their lands and waters.

Example

In Māori culture in **New Zealand** (Aotearoa), there are a number of traditional management tools applied to meet resource management needs and expectations, in particular the expectations of the broader tribal interests. One such 'tool' is known as '*rahui*' and is a form of restriction applied to a resource to ensure the safety of both the resource and community which interacts with it. A *rahui* is generally understood to be a temporary restriction applied by the appropriate tribal representative/s and overseen by the tribal members collectively. It is considered temporary as compared to a full restriction or '*tapu*' which can be placed indefinitely for more apparent reasons.

An example of a temporary *rahui* can be from its application over several hours or days to coincide with a search activity for lost person/s or the short-term approach to protect an area from encroachment during a vulnerable time. Some rahui are regularly applied e.g. as a seasonal tool during the breeding season of key fish or other species. *Rahui* are also applied in many instances to protect natural species or geographical features from exploitation, for example when the harvest season for a key shellfish species is considered to negatively affect its population a *rahui* can be applied to restrict access and/or collection.

Rahui is a traditional tool with very diverse application and effects. Recently a 10-year *rahui* was placed on a wilderness area in the Tongariro region of Central New Zealand, which had suffered a devastating wildfire. The timeframe was considered by the tribal interests as appropriate to ensure full recovery of all species living in that wilderness, including the seasonal species feeding in the area and unseen biota such as insects and others.

Every *rahui* is unique. They carry the cultural nuances of their tribal association. They also need to be valued by the wider community, not only Māori. It it seems that often they are generally respected, primarily through good education programmes with the public.

Participants highlighted that many Indigenous Peoples and local communities are working on revitalizing biocultural systems, including their knowledge systems and food systems, to counteract the damage and decay that have been caused by social changes and outside pressures. Participants noted that if biocultural systems are the focus of restoration, and ILK systems are engaged to sustain these, then restoration of human wellbeing, culture and biodiversity can support each other.

A participant also noted that many communities actively build connectivity, for example in Mexico, communities go to their borders to discuss fire risks or other issues that run across their lands with neighbouring communities. Many communities also recognize the importance of watersheds, as the sources of water that sustain life, including downstream crops and communities. They may therefore engage in specific activities to protect watersheds or to reduce impacts in these areas.

Overall, participants emphasized the crucial importance of customary governance systems for guiding community decision-making around uses of and relationships with landscapes and seascapes, as shown in the examples below.

Moreover, participants highlighted that ILK is not static, and nor are Indigenous and local approaches to land management and spatial planning. These systems have always adapted to changing conditions and contexts, and remain relevant today even in rapidly changing systems.

Examples

Shyagya is the customary governance system of the Tsum community in **Nepal**, which is based on six non-violence principles (not to set fire to the jungle, no hunting, no trapping, no harvesting of honey, no harm to living beings, not to sell cattle for meat) and five precepts (*Panchasheel*) of the Buddhist philosophy. This allows Indigenous Peoples and animals to live in harmony without fear. It has now been formally recognized within the legal system of Nepal, as the Shyagya Conservation Act 2022, in recognition of its effectiveness for managing relationships between people and other aspects of nature (see section 3.8 below for discussion of the political context).

Mukkuvar Fishers in South **India** engage in small scale fishing activities and use Indigenous technologies. Different approaches to fishing are allowed in different areas depending on the season. They resist imported technologies like bottom trawling and ring or purse seines, and often only hook and line fishing is allowed. Fishing expeditions are planned according to the seasonal weather conditions, and the abundance and seasonal availability of fish stocks. Other fish species and marine habitats are left untouched for regeneration and reproduction. Overall, they follow the principle of “maximum distribution and minimum disruption” that protects the ecosystem. The system is managed through an informal policing mechanism within the community. However, the situation is changing and they face various pressures and threats.

In some communities in **Mexico**, a communitarian lifestyle means that lands, resources and responsibilities are shared, to ensure sustainable use of resources and conservation. For example, when a community member turns 18, they have to do free collective work, for example “cleaning” the forest to prevent fire. This allows young people to come to know their territory. Moreover, community authorities are chosen by the community based on their experience, knowledge and qualities. As a foundation to this, the territory is seen as a sacred space that sustains the people.

In **Uganda**, pastoralists manage their use of, and movements over, the landscape through a complex customary governance system, which incudes *Akiriket* – a shrine where elders and community authorities conduct traditional consultations. These shrines are always in forests or in rocky areas. Members of the pastoralist community go there rarely – only if there is a very critical issue that requires consultation. There, senior elders and dreamers determine the cause of a problem, and find answers, through rituals including the interpretation of dreams, or the reading of intestines of cows and goats. They also read the water and sky, by pouring water in a bowl and by watching how the flames and smoke of a fire move in the wind. At the homestead level, a council of elders – *Atem* – discusses dreams and plans how to implement decisions made in the *Akriket*, for example migration routes for people and livestock, and selecting animals for hunting, trees for harvesting and watersheds for protection. If they dreamt about degradation and loss of pasture, they will decide what to do about restoration. They may also discuss herbal medicine if there is a disease outbreak. These discussions and decisions are supported by reports from scouts based on mapping of potential migration routes, pastures and water sources. Then the *Atem* will share their knowledge with leaders of the *kraals* (enclosures for livestock), and they will discuss and dialogue. The information is shared with individual shepherds and herders who are moving across the landscape. Sometimes, they travel 150km away from the homesteads, which they may not see for six months or more. As they move, they balance the time they spend in specific locations to reduce degradation. They also gather seeds in the pastures, or seeds of wild fruits, and sew them in some areas to support restoration. Scouts move around, sharing information and mapping routes and water sources. Increasingly, mobile phones and solar-powered charging stations support the sharing of information.

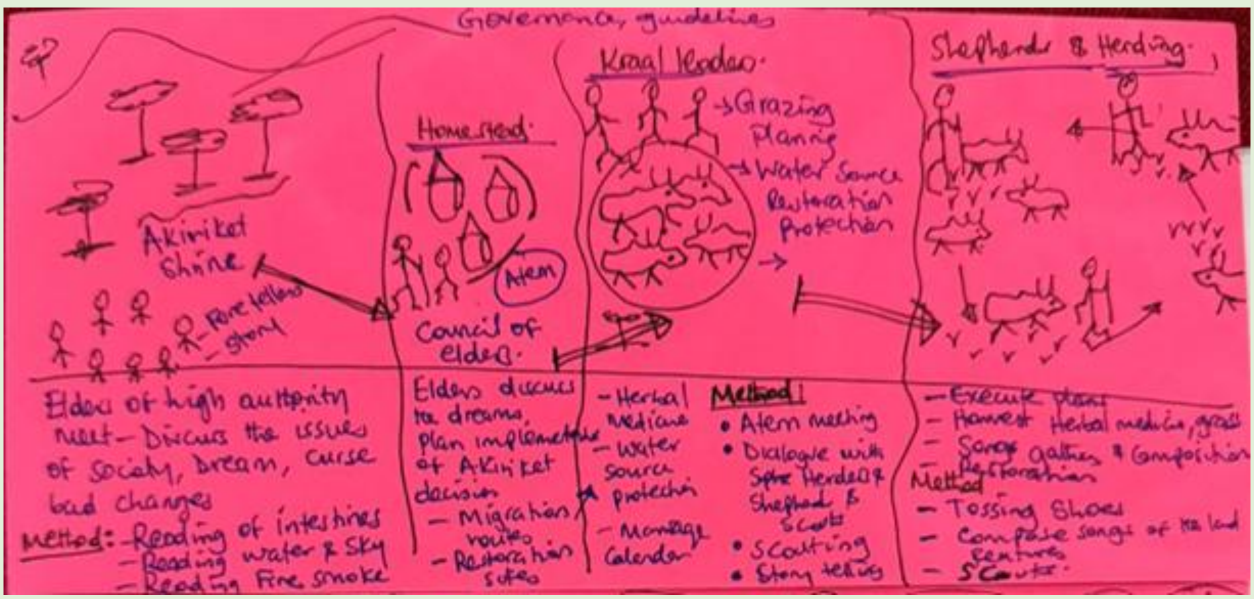

Figure 5: A sketch of the customary governance among pastoralists in Uganda. Information flows from left to right, from the Akiriket (traditional parliament/ shrine or meeting place) to the Atem (council of men and elders) to the kraals (enclosures for livestock) to the individual shepherds and herders (with thanks to Loupa Pius).

3.4 Threats

Participants noted that significant threats exist to spatial planning and management of lands and waters by Indigenous Peoples and local communities. These include changes in value systems, leading to a disconnection of people and nature, for example due to increasing use of technology in farming.

Climate change, including destructive extreme weather events, also threatens the efforts of Indigenous Peoples and local communities to manage their lands and waters and to plan for the future. Issues such as invasive alien species also bring new challenges which further complicate efforts to manage and maintain biodiversity and landscapes.

Overall, participants emphasized that the biggest challenge highlighted by participants is a lack of recognition for Indigenous and local customary governance systems, land rights and knowledge systems, as discussed in section 3.5 below.

3.5 Challenges from externally imposed spatial planning

Participants highlighted that often, in the past and continuing today, formal government-led spatial planning has not included ILK systems, consideration of biocultural systems, or considerations of community rights and wellbeing. This has been the case for zones designated for conservation, and also for industrial development and other uses.

Participants noted that protected areas, or other areas designated for conservation, have often followed a "fortress conservation" approach, in which the only goal was to protect or restore a "pristine" nature. As a result, communities were often forcefully displaced from these areas, with whole villages moved to other areas. Access to the parks was subsequently denied or severely limited, with use of resources forbidden or restricted. This causes many negative impacts on human and Indigenous rights.

Participants emphasized that, considering the description of landscapes and connectivity above, being removed from their lands, waters and resources has serious negative impacts on Indigenous Peoples and local communities, including loss of food security and access to medicines, with consequent impacts on health and wellbeing. Loss of connections to known places, including spiritual sites or burial areas, can also cause great distress. Loss of knowledge systems and identity also occur due to these major ruptures in connectivity between people and landscapes and seascapes, which in turn also further impact community wellbeing and health. Customary governance and management systems can also be severely impacted when the people are cut off from the landscapes and seascapes around which these systems were based. Overall, fortress conservation approaches often lead to the convergent declines or extinction of languages, knowledge and culture.

Meanwhile, tourists and others may be given access to landscapes that local people can no longer visit or use. In some cases, tourists may also visit sacred sites where people were previously forbidden from entering, or which were visited following specific protocols. This causes further distress to communities and increases their feeling of injustice.

Moreover, as described above, Indigenous Peoples and local communities recognize themselves as part of nature, and as part of the balance and harmony of landscapes and seascapes, which they actively seek to maintain. They may therefore see the nature that remains in national parks as unnatural, or out of balance, as it no longer includes people. This is particularly the case when increases in some species lead to increases in human-wildlife conflicts, as is the case in Bardiya, as described below. Other issues, such as uncontrollable wildfires, can also increase when people are removed from a landscape and with them their management practices are also removed.

Participants also noted that concepts and movements that were intended to support Indigenous land rights and control, for example "other effective area-based conservation measures" (OECMs), can also become a challenge for Indigenous Peoples and local communities, as some governments seek to use these concepts and enhance their control of these areas. This can particularly be the case given the drive to enhance areas under protection to reach the 30x30 target of the KMGBF.

Other participants from Nepal also explained that there are also "community managed" areas in the lowlands around Bardiya, and also in the mountains, but that, in practice, community use is often regulated by external actors, permits are needed for all activities, and customary governance is not given authority.

Example of Bardiya National Park in Nepal

Local Tharu Indigenous participants explained that their villages were moved from inside Bardiya National Park, and they were placed in lands on the fringes of the park instead. Access to the park is now forbidden. A community forest has been designated to allow some use of resources, but this is highly regulated, and Tharu people must seek permission to use resources in the area. The *Badhghar* customary governance system of the Tharus is not part of this decision-making and management process.

These restrictions and imposed management systems have had many significant impacts on the local people, as they were removed from their ancestral lands and culturally important areas. They had previously used medicines from the forest, including to treat snake bites and other ailments, but now it is difficult to access these plants. Community members note that this has impacted the knowledge systems in the community, including of medicinal plants, as intergenerational transfer of traditional healing practices and use of medicinal herbs could no longer be ensured.

Moreover, the community has experienced increasing instances of problems with wildlife. The park has been successful in terms of animal numbers, and, in particular, tigers have increased from 121 in 2010 to 355 in 2022. Due to these increases in wildlife numbers, many animals, including tigers, cheetahs, elephants and rhinoceros, frequently leave the park. The creation of a community forest near the village may also be drawing animals closer to the settled areas, while previously Tharu villages would be surrounded by fields. Elephants, rhinoceros and wild pigs can cause significant damage to crops, while tigers, cheetahs and elephants can attack, injure and kill people.

Based on a survey carried out in 2024 of 1,995 households, of which 797 households were from Bardiya, in the Thakurbaba Municipality, 31 cases of loss and damage have been reported by the communities due to wildlife attacks (CIPRED, 2024).[4] Tharu communities report that they face the fear of attacks when engaging in their daily practices like farming, rearing cattle and collecting fodder and firewood from the forests. Indigenous women who are more engaged in these practices and household work are often victims of these attacks. In 2022, in Madhuban Municipality in Bardiya, an Indigenous woman was fatally attacked by a tiger while grazing her cows in the community forest. A week later, another woman of the same village was also attacked by a tiger while working in a field near her home. Similarly, young Tharu children face threats when commuting to their schools, for example a child was killed by an elephant while returning from school.

In response to the frequent attacks by tigers, community members launched a protest demanding protection from wildlife. To diffuse the protest, police fired on the crowd, resulting in the death of a young woman. Her siblings and several other people from the community sustained gunshot injuries.[5]

Overall, these issues are common around many national parks in Nepal, as, in the last three fiscal years, 62 people have died in 104 tiger attacks reported inside national parks and buffer zones in Nepal.[6]

Participants noted that human-wildlife conflict is often related to mismanagement of animals and poor spatial planning. The term “conflict” can imply that the animals are the aggressors, but this is not the case. An example was shared from India, where an elephant walks down the same road every evening, and the local people block the road during that time. As such, space and time can be organized around animals, and spatial planning should consider ILK of animal movements.

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[4] See: https://cipred.org.np/content/331

[5] See: https://www.iucn.org/story/202208/message-chair-spiceh-rights-based-conservation-social-justice

[6] See: https://kathmandupost.com/national/2022/07/26/as-tiger-numbers-rise-experts-stress-protecting-habitats-and-prey-base-reducing-conflict-with-humans

Participants also discussed spatial planning relating to industrial development. A participant explained that in Kerala, India, planners usually do not consider Indigenous knowledge when building roads, and as a result they build the roads over the trails used by elephants. Local people know the trails of the animals, and would have been able to advise the planners to use other routes for the roads. There are other challenges, for example the construction of ports and harbours, without consideration of ILK, which had led to people being killed during monsoon season.

Participants also noted that in terms of community planning, a lack of consideration of ILK can lead to negative results. A participant from Mexico shared the example of a village that was moved by the authorities to a new location, which then flooded. Eventually the community was restored to its original pre-colonial location, which the people had chosen long ago as it was safe from floods.

3.6 Engagement in formal spatial planning processes

Participants highlighted that there are positive examples that demonstrate how Indigenous Peoples and local communities and governments have collaborated around formal planning processes, and where control or management responsibilities have been shared or passed to communities. They emphasized that it will be important for the assessment to highlight these, as they provide models and lessons that could be further developed in different parts of the world.

Example

Bikin National Park in **Russia** was created to protect the Bikin River Valley, which is one of the last large intact tracts of temperate forest in the Northern Hemisphere and habitat for species including Amur tigers. It is also home to the Udege people, who number around 1,450 people across five communities, speaking a local language from the Tungus (Altay) language family. In the past they were nomadic, but now they are more settled into communities. They continue to practice traditional lifestyles, including hunting elk, deer, wild boar and bear, trapping sable, mink and otter, fishing for salmon and other river fish, and gathering of wild plants, including berries, nuts, herbal grass, garlic and fern. With the creation of the park, approximately two million hectares of forest and biodiversity are protected. However, the park is also a good example of co-management and co-creation of a protected area. Traditional knowledge and livelihood practices are recognized and respected, accounting for over 70% of conservation actions. Illegal logging, poaching, and overfishing have been halted in the area. Local Indigenous employment is central and 80% of park staff are from local and Indigenous communities. A sustainable economy is promoted through the non-timber forest products market. It has become a driver of local infrastructure and development, grounded in conservation.

Key mechanisms include FPIC as a mechanism for ongoing dialogue. Creation and maintenance of a multi-stakeholder alliance including green NGOs, academics, legal experts and Udege representatives has also been key, as has joint planning, collective decision-making, and participatory monitoring.

This serves as strong example of what can potentially be achieved. Key lessons learnt shown in this example include:

- High-level negotiations led by Indigenous Peoples, strong written agreements, and the building of strategic coalitions have made this model resilient.
- The red lines of Indigenous consent, benefit-sharing, and participation were respected.
- The inclusion of Indigenous Peoples is not just ethically right. It is also practically effective. Co-management leads to stronger outcomes for both people and planet.
- Success of any sustainability initiative depends on the willingness to reimagine control, rebalance power, and center Indigenous knowledge and leadership.

However, this remains an exclusive case, rather than part of a broader inclusive trend. Furthermore, it cannot be hailed as a complete success, as many challenges remain, and compromises have been made. It is also important to note that a national park is not a replacement for land rights.

A participant from **India** explained that Mukkuvar communities in Kerala often have limited access to policy avenues. They are often considered to be illiterate, and there is a lack of recognition of their knowledge. However, there are two instances of good practices:

- The first Marine Biodiversity Register in India, which incorporated Mukkuvar knowledge, although their contributions were not fully acknowledged; and

- For the assessment of Vizhinjam Bay in the context of port development activities, a publication "Our Beaches, Our Sea"[7] was developed, which aimed to provide a comprehensive analysis of the construction's impact, the disruption of livelihoods, and financial loss incurred by coastal communities, and potential future impacts once the seaport becomes operational. Methods included public consultations and documentation of Mukkuvar knowledge and concerns.

These projects have shown the importance of including coastal Indigenous Peoples like Mukkuvar and their ILK in integrated coastal zone management, environmental impact assessments (EIAs) and any other coastal infrastructure development at the seacoast and in coastal waters. More

[7] https://archtvmnews.com/wp-content/uploads/2024/12/EXECUTIVE-SUMMARY-OF-JPS-REPORT.pdf

broadly, they have shown the need to recognize and respect diverse knowledge systems, and to build inclusive ocean science for effective management of marine spaces.

A participant explained the ways that communities, government and researchers have been working together in **Taiwan, Province of China** around Integrated landscape and seascape approaches. On this small, densely populated island, there are pressures on available space and resources. From a conservation perspective, about 60% of the island is classified as protected areas, including land and sea, and areas outside of protected areas are increasingly being considered as these can also be key for biodiversity, as well as ways to ensure connectivity, for example for leopard cats, which need to move between regions. The government has also conducted a mapping of Indigenous communities, which has shown that they are key to protecting biodiversity, and this understanding has become central to the OECM (other effective area-based conservation measures) strategy of the country.

In **Taiwan, Province of China**, there is also a landscape and seascape shared by two Indigenous groups (Amis and Kabalam), who speak different languages and have different practices and belief systems (Christianity and shamanism). From the 1980s, due to development and urbanization, many youth moved away and elders were left in the tribes. Agricultural practices changed, and began to use more chemicals, which impacted the freshwater and coastal ecosystems. Coastal fishermen noticed damage and changes to the reefs, and noted that the changes were due to issues on the land. As a result, the communities and relevant government agencies came together to develop integrated approaches for land and sea, for development and environmental sustainability, including forestry and organic and ecofriendly food production. Ten years later there is progress in the area, and young people have returned from urban areas and now live permanently in the communities, learning from their elders about how to collect medicinal plants and grow tea, among other activities. There are also positive impacts for ocean ecosystems, and communities are engaged in monitoring the reefs (this is discussed further in the report for the second ILK dialogue for the monitoring assessment). There are now efforts to replicate this across other communities.

A participant from **Uganda** explained the importance of community institutions and networks for managing human-wildlife conflict. Communities form associations to manage their lands and forests resources, which also entails managing wildlife. In some cases, communities and the government have signed memorandums of understanding such as collaborative forest management agreements (CFMA), which lay out roles and responsibilities of the community and government entities. For example, around Moroto District, the Indigenous Tepes people play the role of protecting and conserving Mt. Moroto Forest Reserve from invasion, encroachment and illegal mining activities, and the government’s role is to provide incentives to the local people who are protecting and conserving the forests on behalf of the government. In other cases, community-members are trained and equipped by the National Forest Authority and their own communities, on a voluntary basis, to be able to respond to issues related to human-wildlife conflicts. Communities also patrol their lands to avoid displacement and incursions by other people. These models help to demonstrate that the government can work with communities to protect biodiversity and human wellbeing.

A participant from **Mexico** explained the system of *Ordinamiento Territorial Comunitario*[8] – Community Territorial Land-Use Planning, where the community makes plans with the support of technical experts, and now communities are developing the technical skills themselves. These plans are based on the communitarian lifestyle, in which lands, resources and responsibilities are shared to ensure sustainable use of resources and conservation.

3.7 Methods, protocols and principles

Participants discussed methods that can be used when working with communities, or that communities have developed themselves to elaborate their own planning processes.

Participants emphasized that attention must be given to methods that already exist within ILK systems, which could include ancestral beliefs and spirituality, and to Indigenous methodologies that community members may have developed to further elaborate research in different contexts. In other cases, communities may be weaving their own methods with those of science, merging both to meet their needs and aims.

Several participants discussed the use of mapping, noting that this can be a useful tool for working with Indigenous Peoples and local communities. They noted that many communities have their own mapping traditions, for example in Oceania. Such Indigenous mapping may involve methods that go beyond common ways of mapping, to include diverse values and ways of experiencing the landscape. Meanwhile, more mainstream mapping processes and products can also be useful for working with communities.

Example

A participant shared examples from **Taiwan, Province of China**, where mapping includes maps based on food sources. In some cases, a focus was placed on reducing child malnutrition by focusing on lactating mothers, who often gather food in intertidal and coastal areas. Based on this work, they were able to demonstrate that ships entering these areas are a threat to food security, and they are aiming to create maternal marine reserves.

[8] See: https://dialnet.unirioja.es/servlet/articulo?codigo=5626921

Another participant emphasized the importance of art and other cultural expressions, noting that for many communities, art, songs and other forms of expression are ways of encoding and transmitting knowledge.

In terms of academic disciplines, participants noted the importance of social sciences, anthropology and qualitative approaches, as a counterweight to a common focus on natural sciences and quantitative data. Studies in linguistics can also have value, recognizing that much knowledge is encoded in Indigenous and local languages.

Participants highlighted that through all methodologies, especially where external actors are working with Indigenous Peoples and local communities, intellectual property rights and data sovereignty are key, as laid out in principles such as CARE (collective benefit, authority to control, responsibility and ethics) and OCAP (ownership, control, access and possession), as Indigenous Peoples need to ensure that the use of data and knowledge is to their benefit, and that they have control over how their knowledge is stored and used. FPIC is also an essential foundational principle, as communities should understand the goals, methods and uses of research, and be able to refuse initiatives that do not meet their needs or requirements. They should also be able to refuse to disclose confidential information, for example about the locations or functions of spiritual sites. Systems and protocols may also need to be developed to manage confidential data, so that it can be documented and held by the community, but is not accessible to others.

Participants also emphasized the importance of human-rights based approaches as a foundation for work with Indigenous Peoples and local communities, and in the case of Indigenous Peoples, a focus on the UN Declaration on the Rights of Indigenous Peoples (UNDRIP).

Overall, participants noted that weaving or braiding ILK and science offers many benefits for spatial planning, but that it must be done with appropriate controls and principles, and that processes should be community-led to the fullest extent possible. Participants highlighted that co-production and co-management are very difficult to do well in practice, and should not supersede or replace efforts to better recognize and support ILK methodologies and Indigenous-led processes.

Example

A participant from **Canada** explained how her Anishinaabe community used different methods to establish the site of a new community, including traditional methods such as participation in gathering of roots and berries, preparation of foods, ceremonies, storytelling and visiting community members, alongside methods more common to the social sciences, such as interviews, focus groups and participatory video.[9] The study also engaged community planning and strategic analysis exercises. Maps and mapping were also key. Workshops included a PESCE

[9] For more information see: View of Flooding Hope and Livelihoods: Lake St. Martin First Nation

analysis (political, economic, social, cultural and environmental needs / capitals) to provide a framework from which to structure and highlight community needs and values in relation to a new location for their community.

3.8 Enabling environment

Participants noted that it is essential for the spatial planning and connectivity assessment to explore and highlight key elements for building an appropriate enabling environment that can support the different methods and approaches described above, and that can support the scaling up and scaling out of good examples, including respect for human and Indigenous rights. They noted that there are already some good examples in different countries of the world, as discussed in the sections above and this section, which could be used to demonstrate what can be possible in terms of an enabling environment.

Overall, participants emphasized that it is crucial for governments and other actors to support **better recognition and acceptance of Indigenous and local values, philosophies and conceptualizations about landscapes and seascapes**, as described above in section 3.2. In particular, in the case of the spatial planning and connectivity assessment, Indigenous and local community values, for example biocultural values, need to be promoted in spatial planning and considerations of area-based conservation. This would open space for spatial planning, protected areas, conservation and restoration to address, conserve and restore multiple aspects of biocultural systems, foregrounding human wellbeing and safety (for example around human-wildlife conflicts), and cultural diversity, alongside biodiversity. Remembering and foregrounding ongoing histories of colonialism and dispossession that have taken place in many parts of the world also opens space for spatial planning to become a tool for justice and reconciliation.

Participants noted that **recognition and support for Indigenous ways of undertaking spatial planning** are key. This includes customary systems, as described in section 3.3, and these should be respected and supported even where they are not formally documented. Meanwhile, some communities have developed more formal processes or documents such as biocultural community protocols,[10] *planes de vida* (life plans) or community management plans. These often set out how they want to manage their lands and waters, how their customary governance structures function, and protocols for outside actors to engage with the community. Other communities could also follow this strategy. National governments and other actors could also facilitate the development of such documents, and explore ways to include them in political and legal systems.

[10] For more information, see: https://naturaljustice.org/publication/biocultural-community-protocols/

Participants emphasized that within all of these discussions and activities, **enhanced recognition and respect for ILK systems and customary governance** is key, and governments could aim to develop policy and legal frameworks that support Indigenous Peoples and local communities to manage their own lands and waters following their customary systems. This collaboration between policy and law at the national-level and customary governance and management can provide better outcomes for nature and people, compared to policy and legal frameworks that set national and customary systems against each other. Participants highlighted that learning to devolve power and control can be a key process for national governments.

Participants also highlighted that **thinking through different goals for protected areas, and different ways of designating and managing them**, is another crucial aspect of creating an enabling environment. As noted in the sections above, a historical focus on fortress conservation has in many cases created hardship for Indigenous Peoples and local communities, whilst also failing to conserve biodiversity over broader areas. Exploring other ways of protecting and managing landscapes and seascapes, while recognizing their enmeshed biocultural values and diverse benefits, would allow for the development of a connected system that stretches beyond the existing islands of protected areas, while also supporting Indigenous Peoples' and local communities' cultures, practices, management and governance. Overall, participants noted the need to focus on the quality and multi-functionality of protected area systems, rather than only trying to reach the target of 30% of lands and waters under protection.

Within this, participants highlighted **the importance of the recognition of Indigenous and traditional territories (ITTs)**,[11] as recognized in Target 3 of the KMGBF, noting that this concept offers an Indigenous perspective on the management of lands and waters, and the principles and protocols that should be used to work within these systems. The concept of other effective area-based conservation measures (OECMs) also provides space within discussions around area-based conservation for the recognition of other ways of managing and conserving biodiversity, and biocultural systems. However, as noted above, some participants voiced caution that some governments' drives to designate OECMs within their portfolios of protected areas, stimulated by the target of 30x30 within the KMGBF, risk disempowering Indigenous groups by introducing more top-down management and control. However, they also expressed the hope that the 30x30 target and recognition of diverse ways of exploring area-based conservation could support Indigenous Peoples and local communities. For example, in Mexico there are now proposals to recognize Indigenous territories as protected areas. The Indigenous and Community Conserved Area (ICCA) Consortium also has a global network, conceptualization and registry of Territories

[11] For Guidelines on Indigenous and Traditional Territories in the Context of Biodiversity Conservation, Sustainable Use and Restoration published by the International Indigenous Forum on Biodiversity, see: https://iifb-indigenous.org/iifb-guidelines-on-indigenous-and-traditional-territories-itts/

of Life, which also provides examples and ways of thinking through diverse forms of land management and conservation.[12]

Participants also highlighted that, in all discussions around area-based conservation on Indigenous lands and waters, **respect for human rights and Indigenous rights, as well as the rights of non-humans,** should be a foundation of all processes, decisions and actions, including, crucially, FPIC, and rights to self-determination. **Respect for and consolidation of land tenure** should also be foundations.

Participants also noted that the **crucial role of women** should be highlighted and supported within these systems. In many cases, women play leading roles in maintaining culture, knowledge, traditions and food systems, for example traditions of seed keeping in Peru. However, the crucial role of women is often overlooked within efforts to revitalize and support customary systems.

Examples

The Shyagya customary governance system of the Tsum community in **Nepal**, based on non-violence principles (described in section 3.3 above) has been practiced since time immemorial, with written commitments for the last 102 years, but it was not legally recognized. However, in 2022, the Tsum Nubri Rural Municipality passed the Shyagya Conservation Act 2022, in which the customary institutions became legally recognized. Within this, legitimacy has been given to the Shyagya conservation group and committee, which were formed to implement customary practices which have now become laws. This has been recognized and supported by the Prime Minister of Nepal, who noted that the customary systems have demonstrated possibilities for peace and non-violence. Recognition of the customary systems by local and national governments have demonstrated what can be possible when customary practices are recognized.[13]

A participant shared that in **Mexico**, Indigenous communities have rights over lands and territories, which allowed customary governance systems to be embedded in local-level governance. In Oaxaca, more than 60% of municipalities are governed through these systems.[14]

Participants also highlighted that, where possible, **communities may need to organize themselves and devise strategies** to work within national governance systems, and harness the potential benefits of international agreements, such as the KMGBF.

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[12] See: https://www.iccaregistry.org/en/about/iccas---territories-of-life

[13] See: https://youtu.be/X4ob0gDSUrk?si=u1QNOSM1gl9lMPZk

[14] See: https://www.ieepco.org.mx/sistemas-normativos

For this to occur, **capacity-building may be needed for communities** to better understand their rights and the opportunities within the existing systems, and how to develop and work with tools such as biocultural community protocols. Indigenous organizations can play key roles in this.

Examples

In **Colombia**, the Government has approved the recognition of lands of Indigenous Peoples and has recognized them as environmental authorities. Previously, Indigenous territories were municipalities with the right to do spatial planning and receive funds, but recently this was put into practice with a decree, after a long Indigenous mobilization where they struggled for recognition. Now, Indigenous Peoples are showing how they can function as environmental authorities. This includes the need to build instruments to give legitimacy over their management of their own lands, as well as advocating for broader understanding from the government around the goals and processes used within Indigenous territories, which may not always have biodiversity conservation as an explicit objective, but conservation often occurs anyway as a result of their activities and practices.

Recognizing that most protected areas in the country are on the lands of Indigenous Peoples, Indigenous organizations in **Nepal**, including CIPRED, have been advocating for changes in the laws around protected areas, to include customary governance. This has led to a steady increase in recognition of customary self-governance systems in some areas. However, much more work needs to be done. They are now looking towards the NBSAP process and processes connected to the KMGBF, including the recognition of Indigenous and traditional territories and OECMs as a way of enhancing customary governance and Indigenous control. However, there are risks that OECMs could come to enhance government control and oversight, rather than giving more power to communities. Moreover, the Forest Policy of 2019 indicated that customary management could be recognized as a type of forest management. CIPRED and others have taken this momentum to advocate for recognition of customary governance, in line with the 2015 Nepal constitution. Accessing and compiling disaggregated data is one of the biggest challenges – CIPRED has been focussing on this since 2011 because it is so important to have data to address policy.

A participant from **New Zealand** explained that Māori have many customary tools and methods for conservation. To move forward, some communities have been working on marrying customary approaches with the legal system, so that customary practices can be recognized and protected by law. For example, there is a strip of coastline in the south of the country that has been overfished, and to protect it, communities pushed a mix of customary and legal processes, recognizing that engaging with the formal legal system is the only viable approach in New Zealand. Coastal Māori tribes have also often come together, recognizing that a patchwork of many different systems of control and management may not be feasible, so they have taken a broader approach of collaboration. Overall, Māori have had to prepare themselves to become key players in discussions around management of lands, waters and resources.

Participants also emphasized that **capacity-building among policymakers and other decision-makers** is key, including throughout governance systems from national to local levels, so that the entire formal governance system can become more aligned to working with and supporting Indigenous Peoples and local communities and their customary governance institutions. In particular, participants emphasized that there can be a disconnect between policy at the national level and government officials and institutions at local levels. This becomes a challenge and limitation as it is at these local levels that interactions and collaborations with customary governance systems will often occur. To this end, participants gave examples of networking, for example the Satoyama Network, and peer-to-peer learning between government departments and Indigenous Peoples' organizations. They noted that such activities are in line with Target 20 of the KMBGF "Strengthen Capacity-Building, Technology Transfer, and Scientific and Technical Cooperation for Biodiversity".

4 Next steps

The following next steps will take place after the workshop:

- A report was developed from the dialogue workshop (this report), which was sent to all participants for them to edit, make additions and approve prior to finalization and publication online;
- Using the report as a resource, the authors continued to develop the draft chapters of the assessment;
- Author teams may reach out to participants from Indigenous Peoples and local communities and other members of Indigenous Peoples and local communities, to invite them to be contributing authors; and
- For the assessment of spatial planning and ecological connectivity, another dialogue workshop will be organized in mid-2026 during the external review period for the drafts of the chapters and summary for policymakers, and the assessment will then be considered by the IPBES Plenary at the end of 2027, alongside the monitoring assessment.

Annexes

Annex 1: Agenda

Monday 18 August – the assessment of spatial planning and ecological connectivity	
8.30am-9.00am	Registration
9.00am-9.45am	Opening, introductions: • Opening prayer (Parshuram Choudhari and Ramila Choudhari, community leaders) • Welcome to the participants (Pasang Dolma Sherpa, CIPRED) • Welcome and thanks from the IPBES ILK task force (Nick Rahiri Roskruge, co-chair of the task force) • Introduction of the participants (Vidhya Hirachan and Preity Gurung, CIPRED)
9.45am-10.15am	Introduction to the local area and national context: • Introduction to the community (Tilak Ram Lamsal, Mayor, Thakurbaba Municipality, Bardiya) • Introduction of the community followed by a short video on rights-based conservation (RBC), cases from Bardiya National Park and Suklaphanta National Park (Preity Gurung, CIPRED) • The Nepal NBSAP and Indigenous Peoples (Pooja Shrestha, NIWF)
10.15am-10.45am	Background to the assessments: • Introduction to IPBES and its work with Indigenous and local knowledge (Mariaelena Huambachano) • Context for the workshop and aims, methods and agenda of the dialogue (Peter Bates) • The Kunming-Montreal Global Biodiversity Framework (Q”apaj Conde)
10.45am-11.00am	Refreshment break
11.00am-11.15am	Introduction to the assessment of spatial planning and ecological connectivity – scope, aims, process (David Gonzalez)
11.15am-12.30pm	Discussion: First reflections: • From the IPBES team: what are the aspirations and plans for this assessment relating to ILK? • What are the aspirations and concerns of Indigenous Peoples and members of local communities in relation to spatial planning at different scales? • How could the assessment be useful for Indigenous Peoples and members of local communities? Presentations: • Bikin National Park, Russia (Rodion Sulyandziga) • ILK and marine spatial planning in India (Johnson Jament) • CIPRED experiences with spatial planning, national parks and other effective area-based conservation measures (OECMs) in Nepal (Pasang Dolma Sherpa and Biswash Chepang)
12.30pm-1.30pm	*Lunch*

1.30pm-3.00pm	Discussion: • How do Indigenous Peoples and local communities conceptualize landscapes and seascapes, and conservation, management and restoration?
3.00pm-3.15pm	Refreshment break
3.15pm-5.15pm	Discussion: • How do Indigenous Peoples and local communities engage in landscape and seascape planning, management, restoration and governance, within ILK systems or using ILK systems and mixed methods and tools from science
5.15pm-5.30pm	Preparation for tomorrow's community visit
5.30pm-6.00pm	*Caucus of Indigenous Peoples and local communities*

Tuesday 19 August – community visit	
9.30am	Bus to Community meeting hall, Dalla, Bardiya
10.00am-1.00pm	Welcome to the participants by Tharu traditional dance • Welcome song (Sharmila Tharu) • Welcome to the participants (Chanda Bahadur Tharu, Badghar, community elder) • Objective of the workshop and expectation from the visit (Peter Bates, UNESCO) • Welcome remarks (Tej Bahadur Bhat, Mayor of Madhuban Municipality, Bardiya) • Customary governance systems of Tharu Indigenous communities, "Badghar", how they have been monitoring and managing the resources, benefit sharing mechanism (Parshuram Choudhari) • Questions and answers • Bio Break Issues and concerns of the community in the National Park, buffer zone and community forest (moderated by Pasang Dolma Sherpa, CIPRED) • Human wildlife conflict and compensation (Nayaram Sunar, Bardiya) • Access to the natural resources and ancestral domain (Sumitra Choudhari, Bardiya) • Displacement, eviction and migration (Premwati Rana, Kanchanpur) • Sharing and reflections from the participants • Questions and answers
1.00pm-2.00pm	Lunch
2.00pm-5.00pm	Visiting late Nabina Tharu family member • Introduction of the family • Sharing the story of Nabina • Questions and answers • Tea and snacks Stop to watch local Tharu women fishing and to discuss their knowledge systems
6.30pm-9.00pm	Community dinner

Wednesday 20 August – the assessment of spatial planning and ecological connectivity / the monitoring assessment	
9.00am-9.45am	Reflections on the previous day, planning for the day
9.45am-10.30am	Discussion: • Challenges and benefits for Indigenous Peoples and local communities around spatial planning, e.g., from protected areas, or recognition of "other effective area-based conservation measures"
10.30am-10.45am	Refreshment break
10.45am-12.30pm	Discussion continued: • Challenges and benefits for Indigenous Peoples and local communities around spatial planning, e.g., from protected areas, or recognition of "other effective area-based conservation measures" • Ways forward for Indigenous Peoples and local communities around spatial planning and restoration • What are the principles, methods, tools, capacity needs and policy frameworks that are needed?
12.30pm-1.30pm	*Lunch*
1.30pm-3.00pm	Discussion: • Ways forward for the spatial planning and ecological connectivity assessment – key methods, themes and concepts • What should be the main aims of the assessment? • How can it be useful for Indigenous Peoples and local communities?
3.00pm-3.15pm	Refreshment break
3.15pm-3.45pm	Introduction to the monitoring assessment (Peter Bates and the assessment authors): • Context, aims, methods, timelines, chapters, final product, ILK in the assessment, overview of progress so far, introduction to the ways monitoring is being conceptualized
3.45pm-5.00pm	Discussion / reflections on progress: • The monitoring assessment team's understanding of ILK in this assessment, and how they envision supporting conditions that encourage respectful and beneficial sharing of ancestral knowledge by ILK holders • How can the assessment be useful for Indigenous Peoples and members of local communities? • What are the aspirations and concerns of Indigenous Peoples and members of local communities in relation to monitoring through local, national and global scales?
5.00pm-5.30pm	*Caucus of Indigenous Peoples and local communities*

Thursday 21 August – the monitoring assessment	
9.00am-9.45am	Reflection on the previous day, report back from caucus, plans for today
9.45am-10.45am	Discussion: ILK systems • How do Indigenous Peoples and members of local communities conceptualize and engage in monitoring in their own communities and knowledge systems? What do they think is important to monitor? What is the relationship between monitoring and community decision-making and governance? • How does this differ from "western" scientific methods and priorities? • Reflections on monitoring "nature's contributions to people"? • Developing a figure on monitoring within ILK systems (Paulina Karim)
10.45am-11.00am	Refreshment break
11.00am-12.30pm	Discussion: ILK systems continued
12.30pm-1.30pm	*Lunch*
1.30pm-3.30pm	Presentations and Discussion: Community-based monitoring and information systems (CBMIS) Presentation: • Experiences with CBMIS in Nepal (Biswash Chepang) Discussion: • Sharing the "monitoring typology" (how different types of monitoring are being discussed and categorized in the assessment) • Sharing the conceptualization of CBMIS developed for the assessment • What are the purposes of CBMIS monitoring for communities and their visions for this? (e.g., self-governance, advocacy, territorial claims)? • What elements of ILK can be communicated via CBMIS / other systems? • What may get lost?
3.30pm-3.45pm	Refreshment break
3.45pm-5.25pm	Discussion: from CBMIS to broader collaborative monitoring platforms • In which ways are communities and their organizations interested in co-developing monitoring initiatives? What would that look like? • Can different community initiatives be linked and networked? How? • What are the benefits to Indigenous Peoples and local communities from collaborating on monitoring systems at different scales? • What are the issues and challenges with working on broader monitoring systems at different scales? What are the limitations? What gets lost? • Navigating the relationship between ILK and science in monitoring systems: perspectives on weaving Indigenous and scientific methods respectfully without compromising ILK
5.25pm-5.30pm	Closing of day
5.30pm-6.00pm	*Caucus of Indigenous Peoples and local communities*

Friday 22 August – the monitoring assessment	
9.00am-9.15am	Reflections on previous day, planning for final day
9.15am-10.45am	Discussion: Principles and processes • Strengthening CBMIS/monitoring systems: what principles ensure community control, data sovereignty, and alignment with local values and worldviews? (e.g., including FPIC or the CARE principles) • Participation, agency and decision-making: how do Indigenous Peoples and local communities ensure genuine participation and leadership in monitoring efforts? What does meaningful agency look like? How does this compare to the typologies presented earlier (reflect back and discuss typologies table) • How can tensions between local monitoring needs and global monitoring agendas be addressed? What are the specific risks associated with sharing information, and / or aggregating monitoring information?
10.45am-11.00am	Refreshment break
11.00am-12.30pm	Discussion: How to co-develop monitoring systems / capacity-building and support needs • How can monitoring initiatives be designed so they support Indigenous Peoples and local communities, through capacity-building or support for local governance, etc? • What are the policy needs of Indigenous Peoples and local communities in the context of monitoring? What policies provide strong enabling conditions to support involvement, contributions and leadership in monitoring? • What are the needs in relation to financial support for monitoring, particularly in relation to ensuring the longevity of monitoring initiatives, and dedicated support for developing monitoring in collaboration with Indigenous Peoples and local communities or with ILK?
12.30pm-1.30pm	*Lunch*
1.30pm-2.00pm	*Caucus of Indigenous Peoples and local communities*
2.00pm-2.30pm	Report back from caucus of Indigenous Peoples and local communities
2.30pm-3.30pm	Overview discussion: • What should be the main aims of the assessments? • How can they be useful for Indigenous Peoples and local communities? • How should ILK be acknowledged and protected in global assessments?
3.30pm-3.45pm	Refreshment break
3.45pm-4.15pm	Overview discussion continued
4.15pm-4.30pm	Next steps and participation in the assessments: Timelines for collaboration, communication and dialogue throughout the assessment process
4.30pm-5.00pm	Closing remarks • Participants' reflections • UNESCO • Vidhya Hirachan, CIPRED Closing prayer

Annex 2: FPIC document

Free, Prior and Informed Consent

The individuals whose names are listed at the end of this document agreed during the dialogue workshop to follow the principles and steps laid out in this document.

Background

Within the framework of the UN Declaration on the Rights of Indigenous Peoples (UNDRIP), principles of Free, Prior and Informed Consent (FPIC) apply to research or knowledge-related interactions between Indigenous Peoples and outsiders (including researchers, scientists, journalists). Given that the dialogue process includes discussion of Indigenous and local knowledge of biodiversity and ecosystems, there may be information which the knowledge holders or their organizations or respective communities consider sensitive, private, or holding value for themselves which they do not want to share in the public domain through publications or other media without formal consent.

Principles

The dialogue will be built on equal sharing and joint learning across knowledge systems and cultures. The aim is to create an environment where people feel comfortable and able to speak on equal terms, which is an important precondition for true dialogue.

To achieve these aims, the following goals are emphasized:

- Equality of all participants and absence of coercive influence
- Listening with empathy and seeking to understand each other's viewpoints
- Accurate and empathetic communication
- Bringing assumptions into the open

If participants feel that the above goals are not being achieved at any point during IPBES activities, participants are asked to bring this to the attention of the organizers of the activity, or the IPBES technical support unit on ILK, at: ilk.tsu.ipbes@unesco.org.

Sharing knowledge and respecting FPIC

To ensure that knowledge is shared in appropriate ways during dialogue workshops and other IPBES activities, and that information and materials produced after these activities are used in ways that respect FPIC, the following was put forward:

1. Guardianship – participants who represent organizations and communities

- Principles of guardianship will be discussed with IPLC participants at the beginning of IPBES activities.
- Participants who represent organizations or communities will act as the guardians of the use of the knowledge and materials from their respective organizations or communities that is shared before, during or after the workshop. Any use of their organizations' or communities' knowledge will be discussed and approved by the guardians, as legitimate representatives of their organizations or communities. Guardians are expected to contact their respective organizations and communities when they need advice. Guardians are also expected to seek consent from their organizations or communities when they consider that this is required, keeping in mind that sharing details of their community's knowledge can potentially have negative consequences, for example sharing the locations and uses of medicinal plants.

2. FPIC rights during dialogue workshops and other activities

- The FPIC rights of the Indigenous Peoples participating in dialogue workshops or other activities will be discussed prior to the beginning of the activity, until participants feel comfortable and well informed about their rights and the process, including the eventual planned use and distribution of information. This discussion may be revisited during the activity, and will be revisited at the end of dialogue workshops once participants have engaged in the dialogue process.
- Participants do not have to answer any questions that they do not want to answer, and do not need to participate in any part of an activity in which they do not wish to participate.
- At any point, any participant can decide that they do not want particular information to be documented or shared outside of the activity. Participants will inform organizers and other participants of this. Organizers and participants will ensure that the information is not recorded. Participants can also request that the information is only recorded as a general statement attributed to a region or country, rather than to a specific community.
- Permission for photographs must be agreed prior to photos being taken and participants have the right not to be photographed. Organizers will take note of this.

3. After the activity

- Permission will be obtained before any photograph of a participant is used or distributed in any form.
- Permission will be obtained before any list of participants is used or distributed in any form.

- Participants maintain intellectual property rights over all information collected from them about themselves or their communities, including photographs. Their intellectual property rights should be protected, pursuant to applicable laws.
- Copies of all information collected will be provided to the participants for approval.
- The information collected during the activity will not be used by IPBES for any purposes other than those for which consent has been granted, unless permission is sought and given by participants. Participants note that IPBES does not have control over how others may use its publicly available materials that may contain ILK.
- Participants can decline to consent or withdraw their knowledge or information from the process at any time, and records of that information will be deleted if requested by the participant. Participants should however be aware that once an assessment is published it cannot be changed, and information incorporated into the assessment cannot therefore be withdrawn from the assessment after this point.
- Participants have the opportunity of reviewing and commenting upon the final product during the draft review period, and a dialogue workshop will be organized to support this, bearing in mind that responsibility for the final product rests exclusively with the authors.

Annex 3: Participants of the dialogue workshop

Name	Activity/Organization/Theme
Opening Ceremony – Community Field Visit	
Sharmila Tharu	Song
Chandra Bahadur Tharu	Badhghar customary institution
Nayaram Sunar	Human wildlife Conflict
Sumitra Chaudhary	Access to resources
Premwoti Rana	Displacement/migration

List of Community Consultation Participants	
Nayaram Sonar	Buffer Zone Community Forestry User Groups
Maghuram Chaudhary	Badhghar customary institution
Chandra Bahadur Tharu	Badhghar customary institution
Sumitra Chaudhary Bika	Women's Group
Shristi Chaudhary	Women's Group
Govinda Chaudhary	Youth Group
Amin Chaudhary	Buffer Zone-Community Forest User Committee
Mangal Tharu	Buffer Zone-Community Forest User Committee
Hemanta Acharya	Buffer Zone-Community Forest User Committee
Lila Tharu	Tharu Community Member
Asmita Tharu	Tharu Community Member
Kunti Rana Magar	Tharu Community Member
Jaggu Shahi	Tharu Community Member
Tej Bahadur Bhatt	Mayor, Madhuban Municipality
Mek Bahadur Gurung	NEFIN District Coordination Councils
Saraswoti Rana	Rana Tharu Community Member
Premwoti Rana	Rana Tharu Community Member
Tilak Ram Lamsal	Mayor, Thakur Baba Municipality

National Participants	
Kamal Kumar Rai	Society for Wetland Biodiversity Conservation / former IPBES ILK task force member
Roshani Limboo	Coordinator of the ICCA Youth Network Nepal
Pooja Shrestha	National Indigenous Women Forum Nepal

International participants	
Myrle Ballard	Canada / IPBES ILK task force / University of Calgary
Q"apaj Conde	Bolivia / IPBES ILK task force / CBD secretariat
Yesenia Hernandez	Mexico / IPBES ILK task force

Mariaelena Huambachano	Peru / IPBES ILK task force / Syracuse University
Johnson Jament	India / School of Global Studies, University of Sussex
Paulina Karim	Russia / IPBES ILK task force / National Dong Hwa University
Loupa Pius	Uganda / IPBES ILK task force / KAYESE Community Organisation
Rodion Sulyandziga	Russia / Center for Support of Indigenous Peoples of the North (CSIPN)
Nick Rahiri Roskruge	New Zealand / Co-chair of the IPBES ILK task force / Tahuri Whenua Māori horticultural collective

IPBES assessment authors	
Gaby Lichtenstein	Chapter 1 – Monitoring assessment
Patrick O'Farrell	Chapter 2 – Monitoring assessment
Pablo de la Cruz	Chapter 3 – Monitoring assessment
Olga Lukyanova	Chapter 1 – Assessment of spatial planning
Ehsan Pashanejad	Chapter 2 – Assessment of spatial planning
Medani Bhandari	Chapter 5 – Assessment of spatial planning

CIPRED Participants	
Vidhya Hirachan	Board member
Pasang Dolma Sherpa	Director
Biswash Chepang	Programme Coordinator
Preity Gurung	Programme Officer
Bhabin Rai	Programme Assistant
Narendra Lama	Field Coordinator
Parashuram Chaudhary	Community Focal Person
Ramila Chaudhary	Field Officer

IPBES and UNESCO secretariats	
Peter Bates	IPBES technical support unit on ILK / UNESCO
Claire Bracegirdle	IPBES technical support unit on ILK / UNESCO
David Gonzalez	IPBES secretariat
Barsha Lekhi	UNESCO office in Kathmandu